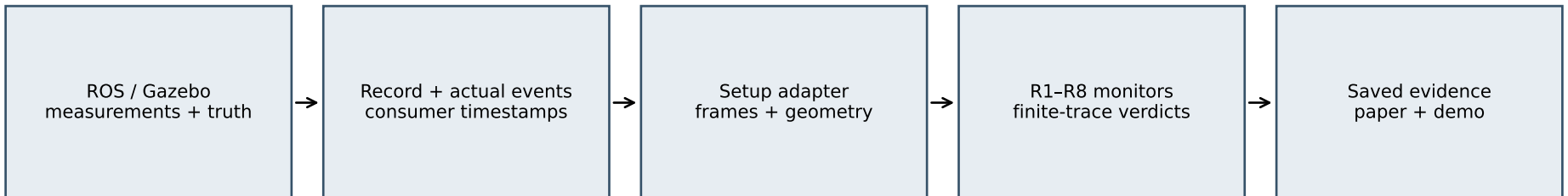
ROS / Gazebo
measurements + truth
Record + actual events
consumer timestamps
Setup adapter
frames + geometry
R1–R8 monitors
finite-trace verdicts
Saved evidence
paper + demo